Sensor Technologies for a Formula Race Car

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Abstract*— Autonomous Formula Student racecars require a reliable sensing and computing architecture to perceive the racing environment, estimate vehicle motion, determine position and execute steering and braking commands. The performance of an autonomous system depends not only on the individual sensors but also on their calibration, synchronization, integration, and ability to operate under uncertain conditions. This report examines the principles, applications, specifications, limitations and procurement considerations of selected sensing and computing technologies, including the Hesai Pandar40P LiDAR, VectorNav VN-200 inertial navigation system, Stereolabs ZED 2i stereo camera, NVIDIA Jetson AGX Orin computing platform, wheel-speed sensors, accelerometers, magnetometers and optical encoders.

Keywords*— Autonomous racing, LiDAR, inertial navigation, stereo vision, sensor fusion, Formula cars

I. Introduction

Sensors form the foundation of an autonomous Formula Student race car because they provide the information required for environmental perception, localization, vehicle-state estimation and feedback control. Unlike a manually driven vehicle, an autonomous race car must identify track boundaries, detect cones and obstacles, estimate its position and orientation, and continuously determine the appropriate steering and braking commands without relying on a human driver. These functions require measurements from several complementary sensing technologies.

Cameras provide visual information, LiDAR measures distances to surrounding objects, inertial sensors measure vehicle motion, GNSS provides satellite-based positioning, and wheel-speed sensors measure wheel rotation. Their outputs are processed by an onboard computing platform and used by perception, planning and control algorithms.

II. Perception Sensors

A. Hesai Pandar40P LiDAR

1) Working Principle

Light Detection and Ranging (LiDAR) measures distance by transmitting laser pulses and detecting their reflections. The Hesai Pandar40P is a rotating, 40-channel three-dimensional LiDAR that generates a point cloud representing surrounding objects. Its published specifications include a 360° horizontal field of view, a 40° vertical field of view, and selectable scanning rates of 10 or 20 Hz. Its maximum specified range can reach approximately 200 m under suitable target-reflectivity and operating conditions; practical performance depends on the target and environment.

The distance to a detected object is calculated using the time-of-flight principle. Combining the measured distance with the laser's direction produces three-dimensional points. These points can be processed into clusters and geometric features for object detection and mapping.

2) Applications

In a Formula Student Driverless race car, the Pandar40P can identify cones marking track boundaries, estimate their relative positions, and support track reconstruction. The resulting cone map can be used by planning algorithms to generate a feasible trajectory. LiDAR measurements may also help identify obstacles and contribute to localization when combined with inertial and other sensor data.

3) Limitations

LiDAR performance can deteriorate when surfaces produce weak reflections, objects are occluded, or the sensor window is contaminated. The rotating mechanism also requires a rigid mount that minimizes vibration and preserves the sensor's alignment. Integration requires suitable Ethernet connectivity, compatible data-processing software, coordinate-frame calibration, and consistent timestamps. The LiDAR should be mounted where the vehicle structure does not obstruct its field of view.

B. Stereo Vision and ZED 2i Camera

1) Working Principle

The ZED 2i is a stereo camera that estimates depth by comparing synchronized images captured from two cameras separated by a known baseline. Its published specifications include a 120 mm baseline and a built-in IMU; the precise camera configuration depends on the selected model. An object appears at slightly different horizontal positions in the left and right images. Calibration is necessary to determine the cameras' intrinsic parameters and their relative alignment.

2) Applications

The ZED 2i can provide both visual information and depth estimates for cones, obstacles, and other track features. Computer-vision algorithms can detect and classify objects, while the associated depth information helps estimate their positions relative to the camera. These outputs can be combined with LiDAR and navigation data to improve environmental perception and support track reconstruction.

3) Limitations

Stereo depth estimation can become less reliable on textureless surfaces, in poor lighting, in glare, or when motion blur affects the images. Depth uncertainty also increases as disparity becomes small at longer distances. The camera must be mounted securely, calibrated relative to the vehicle and other sensors, and connected through a supported high-bandwidth interface. Its timestamped image and depth outputs must be aligned with LiDAR and navigation measurements before sensor fusion.

C. Monocular Camera

1) Working Principle

A monocular camera captures images using a single image sensor. It provides two-dimensional visual information but does not directly measure depth through stereo disparity. Depth must instead be inferred using visual cues, known object dimensions, camera motion, learned models, or additional sensor information. These estimates depend on the assumptions and algorithms used.

2) Applications

A monocular camera can support cone classification, object detection, visual tracking, and recognition of track features. It may be used independently for visual perception or alongside LiDAR to combine image-based object recognition with direct range measurements. A single camera stream from a stereo-camera system can also be processed as a monocular input.

3) Limitations

Monocular depth estimates may be ambiguous because different scenes can produce similar images. Shadows, glare, changing illumination, occlusion, and motion blur can affect detection performance. The system should therefore assess detection confidence and use other sensors where reliable distance information is required.

III. Localization and Vehicle-State Estimation

A. VectorNav VN-200 GNSS/INS

1) Working Principle

The VectorNav VN-200 is a GNSS-aided inertial navigation system that combines inertial measurements, magnetic-field measurements, and satellite-navigation data to estimate position, velocity, and orientation [3]. Its accelerometers measure specific force, its gyroscopes measure angular velocity, and its magnetometer measures magnetic-field components. The navigation solution combines these measurements with GNSS observations to constrain errors that would otherwise accumulate during inertial integration.

2) Applications

The VN-200 can provide navigation and orientation estimates to support trajectory tracking and vehicle-state estimation. Its measurements may be combined with wheel-speed, camera, and LiDAR observations to improve the estimate of vehicle motion. High-rate inertial measurements are useful during brief GNSS interruptions, although they cannot prevent inertial drift indefinitely without external corrections.

3) Limitations and Installation

Navigation performance depends on sensor configuration, calibration, GNSS signal quality, mounting alignment, and environmental conditions. Satellite obstruction and multipath can degrade position estimates, while magnetic interference can affect heading. The unit should be mounted rigidly, its orientation relative to the vehicle should be configured correctly, and GNSS antenna placement and lever-arm offsets should be considered.

B. IMU and INS Fundamentals

1) Working Principle

An Inertial Measurement Unit (IMU) measures angular velocity using gyroscopes and specific force using accelerometers. Some IMUs also include magnetometers. An Inertial Navigation System (INS) processes inertial measurements through navigation algorithms to estimate attitude, velocity, and position. An IMU is therefore a measurement device, whereas an INS includes the processing required to derive a navigation solution.

2) Applications

IMUs support vehicle-motion estimation during acceleration, braking, and cornering. Their high-rate measurements can be combined with GNSS, wheel-speed, camera, and LiDAR data to produce a more consistent estimate of vehicle state. This information allows the control system to compare the vehicle's measured movement with its intended trajectory.

3) Limitations

Small measurement biases and noise accumulate when inertial data is integrated over time, causing errors in estimated velocity and position. Vibration, temperature changes, poor mounting, and incorrect coordinate alignment can further reduce accuracy. External measurements and

sensor-fusion algorithms are therefore important for limiting drift and maintaining a reliable navigation estimate.

C. Accelerometer

1) Working Principle

An accelerometer measures specific force along one or more axes. Many accelerometers use a microscopic proof mass whose movement changes an electrical property inside the device. In a racecar, accelerometer readings help estimate longitudinal acceleration during braking, lateral acceleration during cornering, and vibration caused by track irregularities. Measurement quality depends on sensor bias, noise, temperature, range, and mounting alignment.

2) Types

- **Capacitive MEMS accelerometers:** Detect movement of a proof mass through changes in capacitance. Their compact size and low power consumption make them common in automotive IMUs.
- **Piezoresistive accelerometers:** Measure resistance changes caused by mechanical deformation and can be suitable for shock and high-acceleration measurements.
- **Piezoelectric accelerometers:** Generate electrical charge under mechanical stress and are particularly useful for dynamic vibration measurements. Conventional piezoelectric designs are generally unsuitable for sustained static acceleration measurements.

3) Applications and Limitations

Accelerometers support vehicle-state estimation, braking and cornering analysis, vibration monitoring, and sensor fusion. Combining acceleration measurements with gyroscope, GNSS, and wheel-speed data can improve the estimate of vehicle motion. However, bias, temperature variation, vibration, and saturation can produce inaccurate readings, so the sensor's range and bandwidth must match the application.

D. Magnetometer

1) Working Principle

A magnetometer measures the strength and direction of a magnetic field. Navigation systems may use measurements of the Earth's magnetic field to estimate heading and complement gyroscope and GNSS data. The reliability of this estimate depends on the surrounding magnetic environment and appropriate calibration.

2) Types

- **Hall-effect magnetometers:** Use the Hall effect to measure magnetic fields and can also support magnetic-target detection and position sensing.
- **Magnetoresistive magnetometers:** Detect magnetic fields through changes in a material's electrical resistance.
- **Fluxgate magnetometers:** Use magnetic-core behaviour and electronic excitation to measure magnetic-field components with high sensitivity.

3) Applications and Limitations

Magnetometers can contribute to heading estimation, orientation correction, and magnetic-target detection. However, motors, high-current wiring, and ferromagnetic structures can distort the local magnetic field and introduce heading errors. Their measurements should be calibrated and compared with other navigation sources where possible. A magnetometer alone cannot determine the complete position of a vehicle.

E. GNSS and Navigation Reliability

1) Working Principle

A Global Navigation Satellite System (GNSS) receiver estimates position and velocity from signals transmitted by navigation satellites. It provides an absolute positioning reference that complements the short-term motion information supplied by an INS. Where the receiver and correction infrastructure support it, Real-Time Kinematic (RTK) GNSS can improve positioning accuracy.

2) Applications

GNSS measurements can help constrain inertial drift and provide a global position reference for vehicle localization. Combined GNSS/INS systems can continue producing motion estimates during brief satellite interruptions, using inertial measurements between GNSS updates. The resulting navigation solution can support localization and trajectory tracking.

3) Limitations and Reliability Measures

GNSS accuracy can be reduced by satellite obstruction, reflected signals, interference, and poor antenna placement. Software should monitor fix status, measurement quality, and data age rather than assuming that every position estimate is valid. If navigation uncertainty becomes too high, the vehicle should reduce reliance on the affected estimate and transition to a validated degraded or safe operating state.

IV. Vehicle Feedback Sensors

A. Wheel-Speed Sensors

1) Working Principle

Wheel-speed sensors measure the rotational speed of a wheel by detecting a toothed target or magnetic encoder ring attached to the rotating assembly. Depending on the design, they may use Hall-effect, magneto-resistive, or variable-

reluctance technology. As the target rotates, the sensor produces electrical transitions that a controller processes to calculate wheel speed.

2) Applications and Speed Estimation

If the wheel's angular velocity is (w) and its effective rolling radius is (r), its estimated peripheral speed is v=wr.
The controller can estimate angular velocity from the measured pulse frequency and the known number of pulses per revolution. Wheel-speed data supports vehicle-speed estimation, odometry, and monitoring during acceleration and braking. Comparing measurements from different wheels can also help identify unexpected wheel behaviour.

3) Limitations and Wheel Slip

Wheel peripheral speed may differ from actual ground speed when tyres slip during acceleration or braking. Incorrect sensor alignment, excessive air gaps, damaged targets, electrical noise, and missed pulses can also cause errors. Wheel-speed measurements should therefore be checked against inertial or navigation data rather than treated as a perfect measurement of vehicle speed.

B. Hall-Effect Sensors

1) Working Principle

The Hall effect occurs when a voltage develops across a current-carrying material exposed to a magnetic field. A Hall-effect sensor uses this phenomenon to detect magnetic fields and produce an electrical output. Depending on the device, it may provide a digital switching signal or an analogue measurement.
2) Applications

In a racecar, Hall-effect sensors can detect magnetic encoder targets for wheel-speed measurement and may also be used for motor-position sensing or proximity detection. Certain Hall-effect devices measure magnetic fields generated by electrical current. Their non-contact operation can be useful in applications where mechanical wear should be minimized.

3) Limitations and Selection

Hall-effect sensors generally require a power supply, and their output depends on the sensor type and target arrangement. The target material, magnetic strength, air gap, switching speed, voltage levels, and environmental protection must be checked before selection. A general-purpose Hall-effect component should not be assumed to be a direct replacement for an automotive wheel-speed sensor.

C. Optical Encoders

1) Working Principle and Types

Optical encoders detect changes in light as a coded disk or patterned target moves past an optical sensing element. Incremental encoders produce pulses corresponding to movement, while absolute encoders provide a coded position value. With suitable processing, an encoder can measure rotational speed, displacement, or angular position.

2) Applications

Optical encoders can provide position or rotational feedback for suitable steering actuators, motors, shafts, and test equipment. Their resolution and output can be selected according to the accuracy and response time required by the control system. Where appropriate, encoder measurements can be combined with other feedback signals to check actuator behaviour.

3) Limitations and Integration

Optical encoders can be affected by dust, contamination, moisture, misalignment, and mechanical vibration. Resolution, mounting geometry, output interface, and operating speed must be compatible with the application. For a racecar, their environmental suitability should be assessed before using them near exposed wheels or mechanical assemblies.

D. Steering and Brake Feedback

1) Working Principle

Steering and brake feedback sensors measure the actual response of an actuator or braking system. Steering feedback may include measured steering angle, while brake feedback may include actuator position, pressure, or wheel-speed information, depending on the design. The controller compares the measured response with the requested command to identify tracking errors.

2) Applications

Feedback sensors help the vehicle verify that steering and braking commands are being executed as intended. Steering-angle feedback supports trajectory tracking, while suitable brake feedback can help monitor deceleration and detect unexpected actuator behaviour. These measurements also support system diagnostics and fault detection.

3) Limitations and Safety Requirements

A commanded action does not guarantee that the actuator has achieved the requested result. Faulty sensors, actuator jams, communication loss, and electrical or mechanical failures can cause discrepancies between requested and actual behaviour. The control system should monitor these discrepancies and invoke a defined safe response. Redundancy and fallback behaviour must be designed and validated for the specific vehicle.

V. Computing Platform and Vehicle Control

A. NVIDIA Jetson AGX Orin

1) Architecture and Working Principle

The NVIDIA Jetson AGX Orin is an embedded computing platform designed for robotics and artificial-intelligence workloads. It combines CPU and GPU processing with dedicated acceleration capabilities, allowing perception and autonomous-driving software to run onboard. Actual performance depends on the specific module, power mode, cooling, and software workload.

2) Applications

The Jetson platform can process LiDAR point clouds, camera images, and navigation data while running perception, localization, and planning algorithms. It provides the computational foundation for turning sensor measurements into decisions and control commands. Its interfaces and software support must match the selected sensors and autonomous-driving stack.

3) Limitations

Computational capacity, memory, thermal management, power consumption, and communication bandwidth constrain the workloads that can run reliably. Heavy processing loads or overheating can increase latency or reduce performance. A separate safety mechanism should monitor system health and ensure that a computing failure does not silently lead to unsafe vehicle behaviour.

B. Steer-by-Wire

1) Working Principle

Steer-by-wire transmits steering commands electronically to an actuator rather than relying solely on a conventional mechanical steering input. A controller receives the requested steering angle, commands the actuator, and uses measured steering feedback to determine whether the actual angle follows the target.

2) Applications

In an autonomous racecar, steer-by-wire allows the trajectory-tracking controller to translate a planned path into steering commands. Feedback supports correction of tracking errors and monitoring of steering response. The system can therefore contribute to accurate cornering and trajectory following.

3) Limitations and Safety

The system requires suitable actuators, position feedback, control limits, and fault monitoring. A discrepancy between commanded and measured steering angle must be detected, with a safe response defined for actuator failure, sensor faults, and communication loss. Redundancy and fallback behaviour depend on the vehicle design and applicable competition requirements.

C. Brake-by-Wire

1) Working Principle

Brake-by-wire electronically transmits a braking request to a controller that operates a suitable braking actuator. Depending on the design, the actuator may generate hydraulic pressure or apply braking through an electromechanical mechanism. Relevant feedback sensors monitor the system response.

2) Applications

In a driverless racecar, the controller can request deceleration to follow the planned trajectory, regulate speed, or respond to an emergency. Suitable feedback can help determine whether the braking system is responding as expected. Wheel-speed measurements may also support monitoring of wheel behaviour during braking.

3) Limitations and Safety

Brake-by-wire systems must account for sensor faults, actuator failures, power loss, and communication interruptions. The design needs a validated emergency-stop strategy and a defined response when braking performance cannot be confirmed. These functions must be tested as part of the complete safety architecture, rather than assumed to follow automatically from electronic control.

VI. Reliability, Sensor Fusion and Integration

A. Sensor Fusion

Sensor fusion combines measurements from multiple sources to improve estimates of the vehicle's state and surroundings. For example, LiDAR provides geometric information about cones, cameras contribute visual classification, GNSS provides a position reference, and an IMU captures rapid vehicle motion. An Extended Kalman Filter (EKF) is one method for combining suitable measurements while accounting for uncertainty.

Fusion requires correct calibration, consistent coordinate frames, synchronized timestamps, and appropriate uncertainty estimates. Conflicting measurements should be evaluated rather than blindly averaged. Combining several sensors can improve reliability, but it cannot eliminate common-mode errors or compensate for every sensor failure.

B. Mechanical and Electrical Integration

Sensors should be mounted securely where they have suitable coverage and minimal obstruction. Installation must consider vibration, cable routing, connector retention, heat, dust, moisture, electromagnetic interference, and available mounting space. Electrical supplies and interfaces must meet the selected devices' requirements, while high-current

actuator wiring should be routed to reduce interference with sensitive measurements.

The Jetson AGX Orin, sensor interfaces, actuators, and safety electronics must be treated as one integrated system. Data rates, processing load, thermal management, power consumption, and mechanical packaging should be evaluated together. A component that works on a bench may not provide reliable measurements after installation in a vibrating, electrically noisy racecar.

C. Calibration and Time Synchronization

Calibration establishes the relationship between a sensor's measurements and the vehicle coordinate frame. Camera calibration determines image geometry, stereo calibration establishes the relationship between camera views, and extrinsic calibration aligns the camera, LiDAR, and inertial sensors. Incorrect calibration can shift detected cone positions and distort the estimated track geometry.

Time synchronization is important because the vehicle moves while sensors collect data. Measurements with different timestamps may appear spatially inconsistent even when each sensor is functioning correctly. Where supported, hardware timestamping and suitable synchronization methods should be used, and timestamp consistency should be checked throughout the processing pipeline. Calibration should be verified after sensor mounting changes or maintenance.

D. Fault Detection and Error Recovery

Potential faults include IMU drift, GNSS degradation, missing LiDAR returns, unreliable camera depth, wheel-speed dropouts, magnetic interference, communication timeouts, and computing overload. The software should monitor sensor health, message freshness, measurement plausibility, and disagreement between independent estimates. These checks help distinguish a real vehicle event from a faulty measurement.

When a fault is detected, the system should isolate or reduce reliance on the affected measurement if a validated alternative is available. It should track uncertainty and prevent stale or implausible data from silently influencing control decisions. If the vehicle can no longer estimate its surroundings or motion with sufficient confidence, it should enter a predefined safe state, including stopping when required by the safety design.

VII. Indian Market Sourcing

Component	**Named supplier / source**	**Estimated cost (INR)**
Hesai Pandar40P LiDAR	Hesai Technology — official manufacturer	₹2,50,000–₹4,00,000 (budget estimate)
VectorNav VN-200 GNSS/INS	VectorNav — official product page	₹2,50,000–₹3,50,000 (budget estimate based on listed US-dollar variants)
ZED 2i stereo camera, 4 mm NP	Mouser India — exact product listing	₹57,113.75 listed Mouser India
NVIDIA Jetson AGX Orin 64GB Developer Kit	RoboticsDNA India — product listing	₹1,98,239 including GST (listed; out of stock) RoboticsDNA
Hall-effect sensor IC	Mouser India — electronic components	₹100 (estimated allowance for a basic sensor IC)
Incremental rotary encoder	DigiKey India — CTS 291MV328S1A	₹2,700.24 listed DigiKey
Wheel-speed sensor	HELLA — automotive wheel-speed sensor information	₹1,500 per sensor (budget estimate)
Steering-angle feedback sensor	Bosch Automotive — product and technology information	₹5,000 (budget estimate)

VIII. Conclusion

An autonomous Formula Student racecar depends on coordinated perception, navigation, vehicle-feedback, computing, and actuation systems. The Hesai Pandar40P LiDAR and ZED 2i camera provide complementary environmental information, while the VN-200, IMU, GNSS, and wheel-speed sensors contribute to vehicle-state estimation. The Jetson AGX Orin provides computing and communication capabilities, while steering and braking systems require reliable feedback and validated fault responses.

Reliable operation requires appropriate sensor selection, mounting, calibration, timestamp synchronization, sensor-health monitoring, and testing under realistic conditions. Sensor fusion can improve estimates but cannot eliminate every uncertainty or failure. The system must recognize degraded data and transition to a defined safe state when autonomous operation can no longer be justified.

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